

gets the public key from the Bluetooth-enabled packet and uses the private key to unlock the parcel. If the parcel has been tampered with at any point of time, the customer, on connecting to the Bluetooth portal, receives a warning message, instead of the usual unlock option.

Security is always a challenge

OpeKun took almost a year to take shape from when the idea was conceived. The main challenge was in identification and collaboration, as these require a high level of expertise. What worked for it was combined work by members skilled in different domains.

While there are many smartlocks readily available in the market, in most, the locking systems are attached to doors and, predominantly, use solenoid valves. The solution offered in this product is different, owing to the electromechanical combination put together to specifically address logistics problems on the wheels.

A theft that laid the groundwork

How often have you heard of someone being inspired by a movie or a TV serial, and go on to achieve something huge? Well, here is a case of something similar!

Fig. 2 shows a freight train and a well-secured container with methylamine. Still, calculative minds managed to pull off a heist of a kind, without anyone even realising it. This is a scene from the popular American TV serial *Breaking Bad*, which started it all for Nexshark. It got the makers thinking of how vulnerable the numerous tankers carrying petrol, oil and even water are, which travel around the country on a daily basis.

The team then came up with a hack-proof smartlock combining mechanical and embedded electronics. If implemented in the right manner, this could even solve problems like supplying low-quality products to



Breaking Bad - Season 5

Fig. 2: A scene from American sitcom *Breaking Bad* that inspired the product

farmers or safeguarding bundles of exam papers.

The thoughts—huge and costly

It is only natural to think of what such a solution would truly mean to a vendor. Would the implementation escalate budget costs? Can the delivery agent carry the required number of boxes? The developers plan on providing this solution as a service, meaning that, the cost to the vendor would be a minimal amount for just the transaction.

To make it easy on the person delivering the packet, the developers are looking at using lightweight polyvinyl chloride (PVC) as packaging material. As every box is reusable, the target is to make boxes of standard sizes that can fit most regularly-bought high-end products and transport these safely.

Unlocking the next steps

The team has its plans in place. Having won Makethon organised by IESA and Intel, it is in the process of patenting this stroke of genius.

Thinking beyond the e-commerce industry, the developers are working with pharmaceutical companies to create package containers that auto-lock when the stock reaches its expiry date.

Another area being looked at is an alternate mode of locking for

"We have all heard of the story of Alibaba and the forty thieves. If we look at it closely, we will notice that there was a secure way of opening the caves, the famous Open Sesame. Indian history is rich with such stories. We only need to see the new in the old, and we can make technical marvels. An idea can be simple. Polishing it to get a smooth finish is the key factor."

—Adithya Venkatesh, founder and CEO of Nexshark, shares his view on innovation



transport vehicles carrying milk, petrol and water.

There are also advanced versions of the platform being planned, which would include support for global positioning system (GPS) tracking and measurement of abnormal conditions during transit like shock, impact, temperatures exposed to and the like, using built-in sensors. As one might want to provide authorisation access to a friend/family member, the team is also working on a key sharing setup that would enable transfer of the private key by the customer.

With quality and quantity taking a hit in these uncertain times, the world is looking forward to more such solutions that could help carve out more organised systems. Good luck, team Nexshark! **EFY**